

Instruction for Reproducing the Results of HistoMapFormer: A Swin Transformer-based Model for Global Damage Restoration of Historical Maps with Strong Generalization Capability

ARTICLE HISTORY

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This tutorial, provides a detailed step-by-step instruction for replicating the image quality evaluation process described in this study. We will guide you through the process of setting up the environment and running the HistoMapFormer code to reproduce the experimental results.

1. Purpose

The provided Python script is designed to systematically evaluate the quality of predicted historical maps by comparing them with reference (target) maps. The script computes four common image similarity metrics:

- (1) Mean Squared Error (MSE)
- (2) Root Mean Squared Error (RMSE)
- (3) Mean Absolute Error (MAE)
- (4) Peak Signal-to-Noise Ratio (PSNR)

2. Basic Environment Setup

We'll walk you through every step, from installing dependencies to configuring paper's results.

The hardware used to complete the experiment is as follows:

- **CPU:** Intel Platinum 8358
- **GPU:** NVIDIA A800 80GB PCIe

The software environment is configured as follows:

- **Operating System:** Ubuntu 22.04.4 LTS
- **CUDA Version:** 12.4

We utilize Python for programming, along with the PyTorch framework. Anaconda is used to create and manage the Python environment.

To begin, you'll need to create and activate the Python environment. Run the following commands in your terminal:

```
conda create -n HistoMapFormer python=3.10.8
conda activate HistoMapFormer
```

3. Installing Dependencies

Once the environment is activated, you'll need to install the required dependencies. These include essential libraries such as PyTorch (for deep learning), OpenCV (for image processing), and other supporting packages. Use the following commands to install the dependencies:

```
pip install torch==1.12.1+cu116 torchvision==0.13.1+cu116
torchaudio==0.12.1 --extra-index-url
https://download.pytorch.org/whl/cu116
pip install tqdm==4.67.3
pip install scikit-image==0.25.2
pip install opencv-python==4.8.0.76
pip install pillow==12.2.0
pip install numpy==1.24.0
pip install timm==1.0.26
pip install pywavelets==1.8.0
pip install pytorch-mssim=1.0.0
```

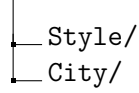
4. Downloading the Code and Dataset

To replicate the experimental results, you'll need to download the code and dataset. The code is available on GitHub, and the dataset can be accessed through Figshare:

- HistoMapFormer GitHub Repository: <https://github.com/giserhh/HistoMapFormer>
- HistoMapFormer Dataset: <https://doi.org/10.6084/m9.figshare.32200377>

After downloading the dataset, place it into the appropriate directories within the project folder(The dataset is strictly organized in the following way, otherwise the script will not read the file correctly.): Data. Our HistoMapFormer project is structured as follows:

```
HistoMapFormer/
├── HistoMapFormer.py .2 requirements.txt # Project dependencies
├── Data/
│   ├── GT/ # Ground Truth
│   │   ├── Class/
│   │   ├── Degree/
│   │   ├── Style/
│   │   └── City/
│   ├── Results/
│   │   ├── Class/
│   │   ├── Class_classify/
│   │   ├── Degree/
│   │   └── Degree_classify/
```



Key Considerations:

- Consistent filenames: The filenames of the GT plot and the results plot must match (including case and extension), or the script will skip the sample.
- Image format: PNG, JPG, JPEG, BMP, TIF, TIFF formats are supported, PNG format is recommended to avoid compression loss.

Our experiment explores the global damage restoration of historical maps from four perspectives: class, degree, style, and city. In the following sections, I will demonstrate how to reproduce the results for each of these aspects.

5. Class(Reproduce the results of Table 1 and Table 2)

All tasks are implemented by modifying the TASK variable in the HistoMapFormer.py script, and the running command is unified as `python HistoMapFormer.py`.

5.1. Reproduce the results of Table 1 HistoMapFormer's results of class

First, open HistoMapFormer.py and modify the TASK variable on line 12:

```
TASK = "Class_classify"
```

After the modification, the corresponding result is shown in the Figure 1:

```

12
13 TASK = "Class_classify"
14
15 TASK_FOLDERS = ["Class_classify", "Class", "Degree_classify", "Degree", "Style", "City"]
16 METHOD_NAMES = ["FFANet", "ChaIR", "CasDyF", "ConvIR", "SADNet", "SwinHistMap"]

```

Figure 1. HistoMapFormer's path in classify(The green text in the figure indicates the content that needs to be modified.)

After the modification, we run the following command:

```
python HistoMapFormer.py
```

The output is shown in the Figure 2.

```

[Class_classify]
      Name MSE      MAE      RMSE      PSNR
-----
    Periodic Banding 0.00004575 0.004797 0.006678 43.614
    Pervasive Aging 0.00002867 0.003538 0.005072 46.265
    Homogeneous Bleaching 0.00003520 0.004455 0.005926 44.554
    Clouded Degradation 0.00003622 0.004053 0.005972 44.539
    Dense Granular Speckling 0.00000410 0.001011 0.002025 53.875
Calculation completed.

```

Figure 2. HistoMapFormer's output in classify.

This is the performance result of HistoMapFormer for global damage restoration on the Table 1:

Table 1. SwinHistMap restoration results for various damage classes

	MSE ↓	MAE ↓	RMSE ↓	PSNR ↑
Periodic Banding	0.00004575	0.004797	0.006678	43.614
Pervasive Aging	0.00002867	0.003538	0.005072	46.265
Homogeneous Bleaching	0.00003520	0.004455	0.005926	44.554
Clouded Degradation	0.00003622	0.004053	0.005972	44.539
Dense Granular Speckling	0.00000410	0.001011	0.002025	53.875

Figure 3. Table 1 in the original manuscript

5.2. Baseline’s results of class

Open HistoMapFormer.py and modify the TASK variable on line 12:

```
TASK = "Class"
```

After the modification, the corresponding result is shown in the Figure 4:

```

13 TASK = "Class"
14
15 TASK_FOLDERS = ["Class_classify", "Class", "Degree_classify", "Degree", "Style", "City"]
16 METHOD_NAMES = ["FFANet", "ChaIR", "CasDyF", "ConvIR", "SADNet", "SwinHistMap"]
17 DEGREE_CLASSES = ["Catastrophic", "Severe", "Slight"]

```

Figure 4. HistoMapFormer’s path in class(The green text in the figure indicates the content that needs to be modified.)

After the modification, we run the following command:

```
python HistoMapFormer.py
```

The output is shown in the Figure 5.

```

[Class]
      Name MSE      MAE      RMSE      PSNR
-----
      FFANet 0.00017084  0.009206  0.011941  39.266
      ChaIR 0.00023785  0.011911  0.013799  38.225
      CasDyF 0.00024189  0.011678  0.014563  37.205
      ConvIR 0.00017485  0.009677  0.012104  39.137
      SADNet 0.00030568  0.013100  0.016605  36.033
      SwinHistMap 0.00002999  0.003571  0.005135  46.569
Calculation completed.

```

Figure 5. HistoMapFormer’s output in class.

This is the performance result of HistoMapFormer for global damage restoration on the class. By incorporating the previous experimental results into a table in Figure 6, Table 2 can be obtained.

6. Degree(Reproduce the results of Table 3 and Table 4)

All tasks are implemented by modifying the TASK variable in the HistoMapFormer.py script, and the running command is unified as python HistoMapFormer.py.

Table 2. Performance evaluation of restoration results on various damage class using MSE, MAE, RMSE and PSNR metrics achieved by different methods.

	MSE ↓	MAE ↓	RMSE ↓	PSNR ↑
FFANet(Qin <i>et al.</i> 2020)	0.00017084	0.009206	0.011941	39.266
ChaIR(Zhang <i>et al.</i> 2022)	0.00023785	0.011911	0.013799	38.225
CasDyF(Wang and He 2024)	0.00024189	0.011678	0.014563	37.205
ConvIR(Cui <i>et al.</i> 2024)	0.00017485	0.009677	0.012104	39.137
SADNet(Niu <i>et al.</i> 2025)	0.00030568	0.0131	0.016605	36.033
SwinHistMap(ours)	0.00002999	0.003571	0.005135	46.569

Figure 6. Table 2 in the original manuscript

6.1. Reproduce the results of Table 3 HistoMapFormer’s results of degree

First, open HistoMapFormer.py and modify the TASK variable on line 12:

```
TASK = "Degree_classify"
```

After the modification, the corresponding result is shown in the Figure 7:

```
12
13 TASK = "Degree_classify"
14
15 TASK_FOLDERS = ["Class_classify", "Class", "Degree_classify", "Degree", "Style", "City"]
16 METHOD_NAMES = ["FFANet", "ChaIR", "CasDyF", "ConvIR", "SADNet", "SwinHistMap"]
17 DEGREE_CLASSES = ["Catastrophic", "Severe", "Slight"]
```

Figure 7. HistoMapFormer’s path in classify(The green text in the figure indicates the content that needs to be modified.)

After the modification, we run the following command:

```
python HistoMapFormer.py
```

The output is shown in the Figure 8. This is the performance result of HistoMap-

```
[Degree_classify]
      Name MSE      MAE      RMSE      PSNR
-----
      Catastrophic 0.00015532 0.008818 0.012407 38.163
      Severe 0.00002980 0.004116 0.005455 45.271
      Slight 0.00001281 0.002569 0.003579 48.929

Calculation completed.
```

Figure 8. HistoMapFormer’s output in classify.

Former for global damage restoration on the Table 3:

6.2. Baseline’s results of degree

Open HistoMapFormer.py and modify the TASK variable on line 12:

```
TASK = "Degree"
```

After the modification, the corresponding result is shown in the Figure 10:

After the modification, we run the following command:

```
python HistoMapFormer.py
```

Table 3. SwinHistMap restoration results for various damage degrees

	MSE ↓	MAE ↓	RMSE ↓	PSNR ↑
Catastrophic	0.00015532	0.008818	0.012407	38.163
Severe	0.00002980	0.004116	0.005455	45.271
Slight	0.00001281	0.002569	0.003579	48.929

Figure 9. Table 3 in the original manuscript

```

12
13 TASK = "Degree"

```

Figure 10. HistoMapFormer’s path in degree(The green text in the figure indicates the content that needs to be modified.)

```

[Degree]

```

Name	MSE	MAE	RMSE	PSNR
FFANet	0.00008298	0.005993	0.008025	43.110
ChaIR	0.00017332	0.010129	0.012347	38.728
CasDyF	0.00008878	0.006222	0.008516	42.301
ConvIR	0.00009054	0.006378	0.008435	42.579
SADNet	0.00017341	0.008659	0.011257	40.477
SwinHistMap	0.00006598	0.005167	0.007147	44.121

```

Calculation completed.

```

Figure 11. HistoMapFormer’s output in degree.

The output is shown in the Figure 11.

This is the performance result of HistoMapFormer for global damage restoration on the degree. By incorporating the previous experimental results into a table in Figure 12, Table 4 can be obtained.

Table 4. Performance evaluation of restoration results on various damage degree using MSE, MAE, RMSE and PSNR metrics achieved by different methods.

	MSE ↓	MAE ↓	RMSE ↓	PSNR ↑
FFANet	<u>0.00008298</u>	<u>0.005993</u>	<u>0.008025</u>	<u>43.110</u>
ChaIR	0.00017332	0.010129	0.012347	38.728
CasDyF	0.00008878	0.006222	0.008516	42.301
ConvIR	0.00009054	0.006378	0.008435	42.579
SADNet	0.00017341	0.008659	0.011257	40.477
SwinHistMap(ours)	0.00006598	0.005167	0.007147	44.121

Figure 12. Table 4 in the original manuscript

7. Style(Reproduce the results of Table 5, Table 6, Table 7 and Table 8)

All tasks are implemented by modifying the TASK variable in the HistoMapFormer.py script, and the running command is unified as python HistoMapFormer.py.

7.1. Baseline's results of Style.

Open HistoMapFormer.py and modify the TASK variable on line 12:

```
TASK = "Style"
```

After the modification, the corresponding result is shown in the Figure 13:

```
12
13 TASK = "Style"
```

Figure 13. HistoMapFormer's path in style(The green text in the figure indicates the content that needs to be modified.)

After the modification, we run the following command:

```
python HistoMapFormer.py
```

The output is shown in the Figure 14.

[Style/1636]					
	Name	MSE	MAE	RMSE	PSNR
	FFANet	0.00004258	0.004944	0.006512	43.744
	ChaIR	0.00006217	0.006355	0.007876	42.083
	CasDyF	0.00011005	0.007399	0.010281	39.933
	ConvIR	0.00013908	0.009452	0.011760	38.617
	SADNet	0.00003614	0.004577	0.005998	44.461
	SwinHistMap	0.00002585	0.003730	0.005074	45.911
[Style/1682]					
	Name	MSE	MAE	RMSE	PSNR
	FFANet	0.00005309	0.005320	0.007265	42.800
	ChaIR	0.00006413	0.006411	0.008006	41.935
	CasDyF	0.00007685	0.006374	0.008742	41.190
	ConvIR	0.00016942	0.010599	0.012933	37.818
	SADNet	0.00004604	0.005108	0.006771	43.405
	SwinHistMap	0.00003167	0.004245	0.005626	44.999
[Style/grid]					
	Name	MSE	MAE	RMSE	PSNR
	FFANet	0.00004452	0.005126	0.006671	43.517
	ChaIR	0.00008598	0.007773	0.009269	40.662
	CasDyF	0.00031651	0.012455	0.017748	35.038
	ConvIR	0.00105152	0.007005	0.032022	30.037
	SADNet	0.00003953	0.004762	0.006286	44.033
	SwinHistMap	0.00002749	0.003933	0.005242	45.612
[Style/youhua]					
	Name	MSE	MAE	RMSE	PSNR
	FFANet	0.00008677	0.007027	0.009315	40.616
	ChaIR	0.00007574	0.006951	0.008703	41.207
	CasDyF	0.00017979	0.009932	0.013409	37.452
	ConvIR	0.00089109	0.021347	0.029851	30.501
	SADNet	0.00007083	0.006407	0.008416	41.498
	SwinHistMap	0.00004387	0.005224	0.006623	43.578

Figure 14. HistoMapFormer's output in style.

This is the performance result of HistoMapFormer for global damage restoration on the degree. By incorporating the previous experimental results into a table in Figure ??, Table 5,6,7 and 8 can be obtained.

Table 5. Performance evaluation of restoration results on symbolic style using MSE, MAE, RMSE and PSNR metrics achieved by different methods.

	MSE ↓	MAE ↓	RMSE ↓	PSNR ↑
FFANet	0.00004258	0.004944	0.006512	43.744
ChaIR	0.00006217	0.006355	0.007876	42.083
CasDyF	0.00011005	0.007399	0.010281	39.933
ConvIR	0.00013908	0.009452	0.01176	38.617
SADNet	<u>0.00003614</u>	<u>0.004577</u>	<u>0.005998</u>	<u>44.461</u>
SwinHistMap(ours)	0.00002585	0.00373	0.005074	45.911

Figure 15. Table 5 in the original manuscript

Table 6. Performance evaluation of restoration results on pictorial style using MSE, MAE, RMSE and PSNR metrics achieved by different methods.

	MSE ↓	MAE ↓	RMSE ↓	PSNR ↑
FFANet	0.00005309	0.00532	0.007265	42.800
ChaIR	0.00006413	0.006411	0.008006	41.935
CasDyF	0.00007685	0.006374	0.008742	41.19
ConvIR	0.00016942	0.010599	0.012933	37.818
SADNet	<u>0.00004604</u>	<u>0.005108</u>	<u>0.006771</u>	<u>43.405</u>
SwinHistMap(ours)	0.00003167	0.004245	0.005626	44.999

Figure 16. Table 6 in the original manuscript

Table 7. Performance evaluation of restoration results on grid style using MSE, MAE, RMSE and PSNR metrics achieved by different methods.

	MSE ↓	MAE ↓	RMSE ↓	PSNR ↑
FFANet	0.00004452	0.005126	0.006671	43.517
ChaIR	0.00008598	0.007773	0.009269	40.662
CasDyF	0.00031651	0.012455	0.017748	35.038
ConvIR	0.00105152	0.007005	0.032022	30.037
SADNet	<u>0.00003953</u>	<u>0.004762</u>	<u>0.006286</u>	<u>44.033</u>
SwinHistMap(ours)	0.00002749	0.003933	0.005242	45.612

Figure 17. Table 7 in the original manuscript

8. City(Reproduce the results of Table 9)

All tasks are implemented by modifying the TASK variable in the HistoMapFormer.py script, and the running command is unified as python HistoMapFormer.py.

Table 8. Performance evaluation of restoration results on Oil painting style using MSE, MAE, RMSE and PSNR metrics achieved by different methods.

	MSE ↓	MAE ↓	RMSE ↓	PSNR ↑
FFANet	0.00008677	0.007027	0.009315	40.616
ChaIR	0.00007574	0.006951	0.008703	41.207
CasDyF	0.00017979	0.009932	0.013409	37.452
ConvIR	0.00089109	0.021347	0.029851	30.501
SADNet	<u>0.00007083</u>	<u>0.006407</u>	<u>0.008416</u>	<u>41.498</u>
SwinHistMap(ours)	0.00004387	0.005224	0.006623	43.578

Figure 18. Table 8 in the original manuscript

8.1. Reproduce the results of Table 9 HistoMapFormer’s results of city

First, open HistoMapFormer.py and modify the TASK variable on line 12:

```
TASK = "City"
```

After the modification, the corresponding result is shown in the Figure 19:

```
13 TASK = "City"
```

Figure 19. HistoMapFormer’s path in city(The green text in the figure indicates the content that needs to be modified.)

After the modification, we run the following command:

```
python HistoMapFormer.py
```

The output is shown in the Figure 20. This is the performance result of HistoMap-

```
[City]
-----
      Name  MSE      MAE      RMSE      PSNR
-----
    GuangZhou 0.00002980  0.004116  0.005455  45.271
    HongKong  0.00003186  0.004363  0.005626  45.025
    Tokyo    0.00032406  0.007860  0.009399  43.713
Calculation completed.
```

Figure 20. HistoMapFormer’s output in classify.

Former for global damage restoration on the Table 9:

Table 9. Performance evaluation of restoration results on various cities using MSE, MAE, RMSE and PSNR metrics.

	MSE ↓	MAE ↓	RMSE ↓	PSNR ↑
GuangZhou	0.00002980	0.004116	0.005455	45.271
HongKong	0.00003186	0.004363	0.005626	45.025
Tokyo	0.00008395	0.005257	0.00661	44.67753

Figure 21. Table 9 in the original manuscript

9. Other Figures

The original images of Figure 8, Figure 9 Figure 10 Figure 11, Figure 12, Figure 13 and Figure 14 in the article can all be viewed and downloaded on the Figshare website: <https://doi.org/10.6084/m9.figshare.32200377>